NamaTravel Final Project

Web API + Entity Framework + SQLite + FlightAPI Integration

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1. Project Overview

NamaTravel is a travel booking web application developed using:

- ASP.NET Core Web API
- Entity Framework Core
- SQLite Database
- FlightAPI (External Flight API)
- HTML, CSS, JavaScript (Frontend)
- Fetch API (AJAX)

The system allows users to:

- View destinations
- Search for flights
- Create bookings
- Update bookings
- Delete bookings
- Store booking data in a database

2. System Architecture

The project follows a 3-Layer Architecture:

Frontend (HTML/CSS/JS)
⬇
Fetch API calls
⬇
ASP.NET Core Web API
⬇
Entity Framework Core

↓
SQLite Database

3. Database Design

Database: SQLite
ORM: Entity Framework Core

Table: Bookings

Fields:

- Id (Primary Key, Auto Increment)
- ClientName (TEXT)
- Destination (TEXT)
- TravelDate (TEXT)
- Travelers (INTEGER)

Entity Framework was used with Code-First approach and migrations.

4. CRUD Operations Implemented

The API provides the following endpoints:

GET /api/bookings
POST /api/bookings
PUT /api/bookings/{id}
DELETE /api/bookings/{id}

All operations were tested using Swagger.

5. External API Integration

The project integrates FlightAPI to retrieve flight information.

The backend controller sends a request to:

https://api.flightapi.io/

The API key is stored securely in appsettings.json.

6. Technologies Used

- ASP.NET Core (.NET 8/10)
- Entity Framework Core
- SQLite

- JavaScript Fetch API
- Swagger
- GitHub

7. How to Run the Project

1. Open backend folder
2. Run:
   dotnet restore
   dotnet ef database update
   dotnet run
3. Open frontend using Live Server
4. Access Swagger at:
   https://localhost:PORT/swagger

8. Conclusion

This project demonstrates:

- Web API development
- Database integration using Entity Framework
- Full CRUD implementation
- Frontend–Backend communication using Fetch API
- External API integration
- Clean layered architecture